```
accuracy and generalization performance.",
"Interestingness": 9,
"Feasibility": 8,
"Novelty": 8,
"novel": true
```

Idea 6/50 - local_vs_global_attention_grokking

```
"Name": "local_vs_global_attention_grokking",
"Title": "Local vs Global Attention: Investigating the Impact of Attention
Scope on Grokking in Algorithmic Learning",
"Experiment": "Modify the DecoderBlock class to support two attention
mechanisms: full (global) attention and local attention with a fixed window
size. Implement these variants and run experiments across all datasets.
Track metrics including time to grokking (99%
validation accuracy, and training loss. Calculate and compare 'attention
entropy' for both mechanisms across tasks to quantify attention focus.
Analyze how the scope of attention (local vs global) affects grokking
behavior and final performance for different algorithmic tasks.",
"Interestingness": 9,
"Feasibility": 8,
"Novelty": 8,
"novel": true
```

Idea 7/50 - input_encoding_grokking

```
"Name": "input_encoding_grokking",
"Title": "Binary vs One-Hot Encoding: Impact on Grokking in Algorithmic
Learning Tasks",
"Experiment": "Modify the AbstractDataset class to support two encoding
schemes: one-hot (current) and binary. Implement binary encoding for
modular arithmetic operations using log2(p) bits, and for permutations
using ceil(log2(k!)) bits to represent each permutation uniquely. Adjust
the Transformer class to accommodate different input sizes. Run experiments
for each encoding scheme across all datasets, tracking metrics such as time
to grokking (99%
loss, and model memory usage. Analyze how different encoding schemes affect
grokking behavior, convergence speed, and final performance for various
algorithmic tasks. Compare the impact of input representations on the
model's ability to learn and generalize across different operations.
Discuss how findings could inform input representation choices in other
machine learning tasks beyond algorithmic learning.",
"Interestingness": 9,
"Feasibility": 8,
"Novelty": 8,
"novel": true
```

Idea 8/50 - curriculum_learning_grokking

```
"Name": "curriculum_learning_grokking",
"Title": "Curriculum Learning in Grokking: The Effect of Structured Example
Progression on Algorithmic Learning",
"Experiment": "Modify the AbstractDataset class to implement a simple
```

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curriculum learning strategy. For modular arithmetic operations, start with
operations involving numbers in the lower half of the range and gradually
introduce larger numbers. For permutations, begin with permutations that
differ from the identity by one swap and progressively increase the number
of swaps. Implement a curriculum scheduler that increases difficulty every
500 steps. Run experiments comparing standard random sampling vs.
curriculum learning across all datasets. Track metrics including time to
grokking (99%
loss. Plot learning trajectories (validation accuracy over time) for both
approaches. Compare attention patterns between curriculum and random
approaches at different stages of training. Analyze how the curriculum
affects the grokking phenomenon across different operations and discuss
implications for training neural networks on algorithmic tasks.",
"Interestingness": 9,
"Feasibility": 8,
"Novelty": 8,
"novel": true
```

Idea 9/50 - weight_init_grokking

```
"Name": "weight_init_grokking",
"Title": "Weight Initialization Strategies and Their Impact on Grokking in
Algorithmic Learning",
"Experiment": "Modify the Transformer class to support three weight
initialization strategies: Xavier/Glorot, Kaiming/He, and random normal (as
baseline). Implement these initialization methods for linear layers and
embeddings. Run experiments across all datasets for each initialization
strategy. Track metrics including time to grokking (99%
accuracy), final validation accuracy, training loss, and gradient norm
during training. Plot learning curves and compare the distribution of
weight values at different stages of training. Analyze the loss landscape
by computing gradient variance as a proxy for local geometry at key points
during training. Compare how different initialization strategies affect the
grokking phenomenon, convergence speed, and final performance across
various algorithmic tasks. Investigate potential correlations between
initial weight distributions, gradient variance characteristics, and the
timing/nature of the grokking transition.",
"Interestingness": 9,
"Feasibility": 9,
"Novelty": 8,
"novel": true
```

Idea 10/50 - task_complexity_grokking

```
"Name": "task_complexity_grokking",
"Title": "Grokking Across Task Complexity: Mapping Neural Network Learning
Dynamics to Algorithmic Difficulty",
"Experiment": "1. Modify the AbstractDataset class to include new
operations of increasing complexity: modular addition, subtraction,
multiplication, and exponentiation. 2. Implement these operations in new
dataset classes. 3. Quantify task complexity using metrics like algebraic
degree and average solution time for humans (estimated). 4. Run experiments
for each operation, tracking metrics such as time to grokking (99%
validation accuracy), final validation accuracy, training loss, and a new
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'complexity-adjusted learning rate' (validation accuracy improvement per
epoch, normalized by task complexity). 5. Plot learning curves and
complexity-adjusted learning rates for each operation. 6. Analyze attention
patterns and hidden state representations at different stages of training
for each operation. 7. Investigate correlations between quantified task
complexity and grokking characteristics (e.g., time to grokking, steepness
of accuracy improvement).",
"Interestingness": 9,
"Feasibility": 8,
"Novelty": 8,
"novel": true
```

Idea 11/50 - regularization_grokking

```
"Name": "regularization_grokking",
"Title": "The Role of Regularization in Grokking: How L2 and Label
Smoothing Affect Algorithmic Learning",
"Experiment": "1. Implement L2 regularization by adding weight decay to the
optimizer. 2. Implement label smoothing in the loss function. 3. Modify the
training function to support these regularization techniques with two
strength levels each (low and high). 4. Run experiments for each
regularization technique and strength across all datasets, including a
baseline without regularization. 5. Track metrics: time to grokking (99%
validation accuracy), final validation accuracy, training loss, and a new
'grokking speed' metric (rate of validation accuracy improvement from 50%
to 90%
for different regularization settings. 7. Analyze how L2 regularization and
label smoothing affect the timing, speed, and nature of grokking across
various algorithmic tasks, comparing against the non-regularized
baseline.",
"Interestingness": 9,
"Feasibility": 9,
"Novelty": 8,
"novel": true
```

Idea 12/50 - grokking_extrapolation

```
"Name": "grokking_extrapolation",
"Title": "Grokking and Extrapolation: Investigating the Limits of
Algorithmic Understanding",
"Experiment": "1. Modify AbstractDataset to create a separate test set with
out-of-distribution examples (e.g., larger numbers for modular arithmetic,
longer permutations). 2. Implement a new evaluation function for the test
set. 3. During training, periodically evaluate on both validation and test
sets. 4. Track metrics: time to grokking on validation set, final
validation accuracy, test set accuracy at grokking point, final test set
accuracy, and 'extrapolation gap'. 5. Implement visualization of test set
performance and extrapolation gap over time, highlighting the grokking
point. 6. Compare extrapolation capabilities across different operations
and model sizes. 7. Analyze attention patterns on test set inputs before
and after grokking. 8. Implement a simple MLP baseline for comparison.",
"Interestingness": 9,
"Feasibility": 8,
"Novelty": 9,
```